

GENERAL CHEMISTRY



Chapter 10 Gases

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10-1 Characteristics of Gases

- ◆ Unlike liquids and solids, gases
 - expand to fill their containers;
 - are highly compressible;
 - have extremely low densities.

- ◆ Vapors are substances that are normally liquids or solids at room temperature.

◆ The gaseous states of three halogens.



Cl_2
greenish
yellow

Br_2
brownish
red

I_2
violet

◆ Most common gases are colorless

- H_2 , O_2 , N_2 , CO and CO_2

TABLE 10.1 Some Common Compounds That Are Gases at Room Temperature

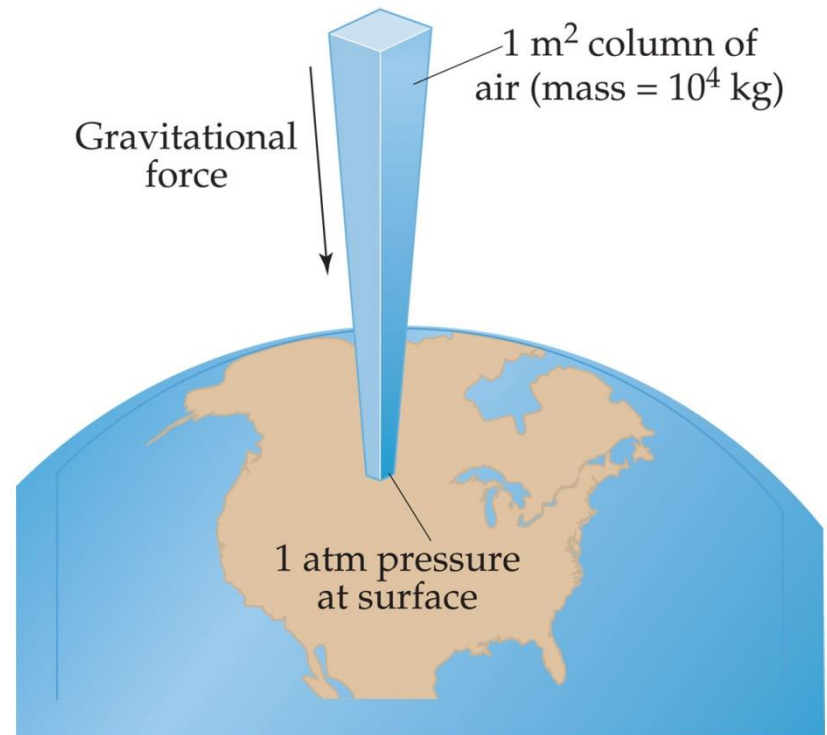
Formula	Name	Characteristics
HCN	Hydrogen cyanide	Very toxic, slight odor of bitter almonds
H ₂ S	Hydrogen sulfide	Very toxic, odor of rotten eggs
CO	Carbon monoxide	Toxic, colorless, odorless
CO ₂	Carbon dioxide	Colorless, odorless
CH ₄	Methane	Colorless, odorless, flammable
C ₂ H ₄	Ethene (Ethylene)	Colorless, ripens fruit
C ₃ H ₈	Propane	Colorless, odorless, bottled gas
N ₂ O	Nitrous oxide	Colorless, sweet odor, laughing gas
NO ₂	Nitrogen dioxide	Toxic, red-brown, irritating odor
NH ₃	Ammonia	Colorless, pungent odor
SO ₂	Sulfur dioxide	Colorless, irritating odor

10-2 Pressure

- ◆ Pressure is the amount of force applied to an area.

$$\text{Pressure (Pa)} = \frac{\text{Force (N)}}{\text{Area (m}^2\text{)}}$$

- ◆ Atmospheric pressure is the weight of air per unit of area.



Units of Pressure

◆ Pascals

- $1 \text{ Pa} = 1 \text{ N/m}^2$

◆ Bar

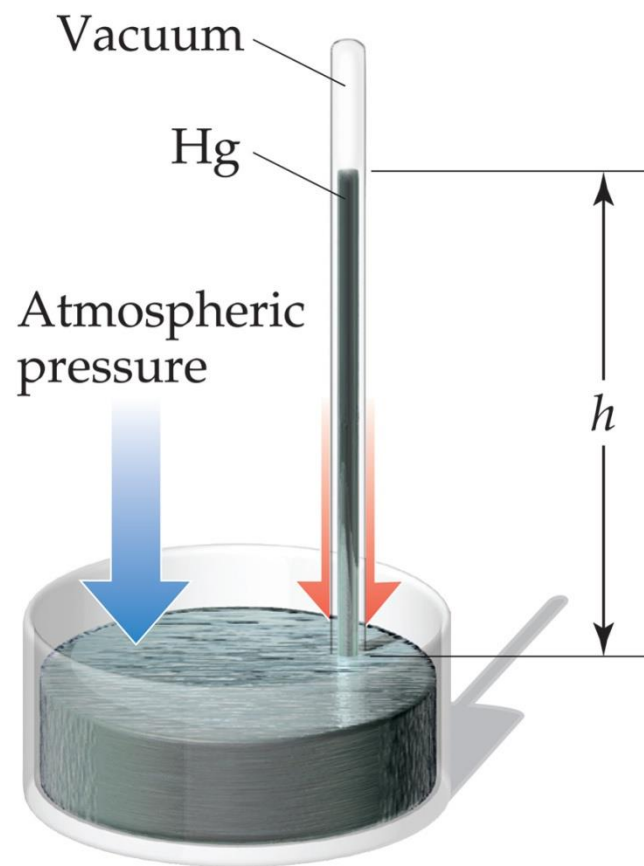
- $1 \text{ bar} = 10^5 \text{ Pa} = 100 \text{ kPa}$

◆ mm Hg or torr

- These units are literally the difference in the heights measured in mm (h) of two connected columns of mercury.

◆ Atmosphere

- $1.00 \text{ atm} = 760 \text{ torr}$



Liquid Pressure

- ◆ The pressure exerted by a liquid depends on:
 - The height of the column of liquid.
 - The density of the column of liquid.

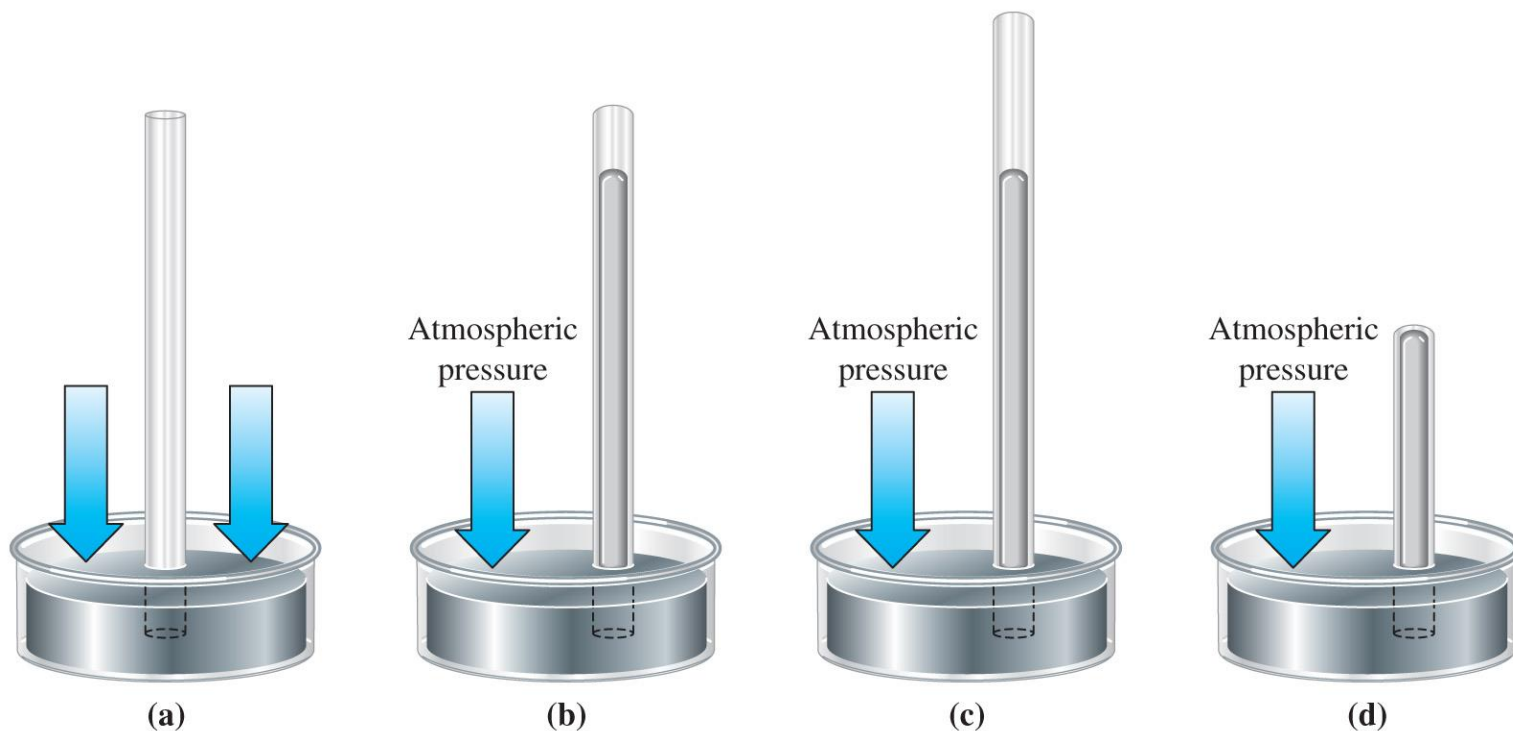


$$P = g \cdot h \cdot d$$

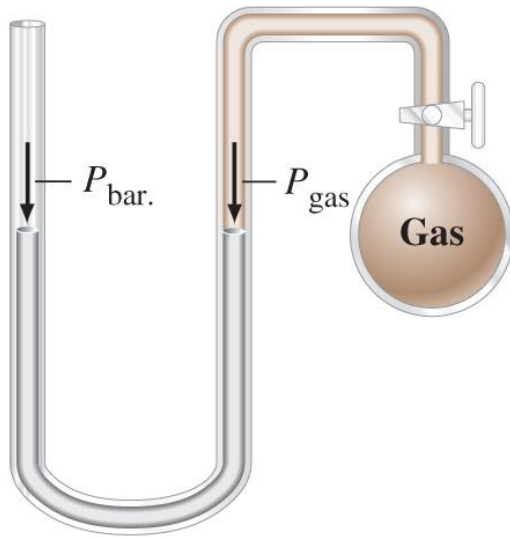
Barometric Pressure

Standard Atmospheric Pressure

1.00 atm, 760 mm Hg, 760 torr, 101.325 kPa, 1.01325 bar

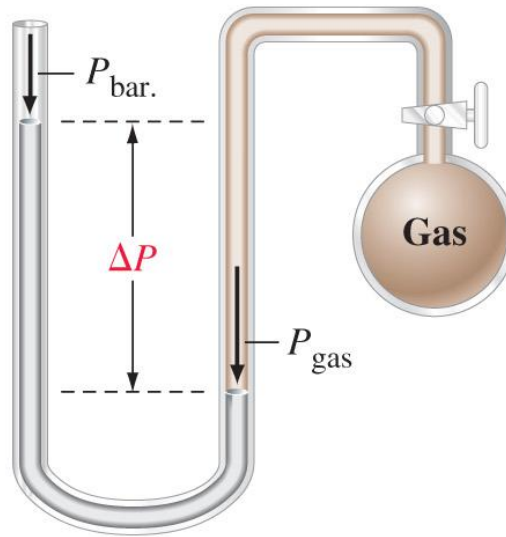


Manometers



$$P_{\text{gas}} = P_{\text{bar.}}$$

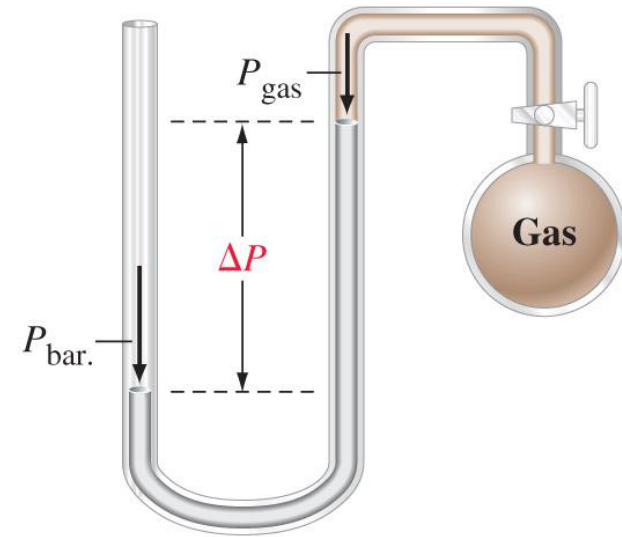
(a) Gas pressure equal to barometric pressure



$$P_{\text{gas}} = P_{\text{bar.}} + \Delta P$$

$(\Delta P > 0)$

(b) Gas pressure greater than barometric pressure



$$P_{\text{gas}} = P_{\text{bar.}} + \Delta P$$

$(\Delta P < 0)$

(c) Gas pressure less than barometric pressure

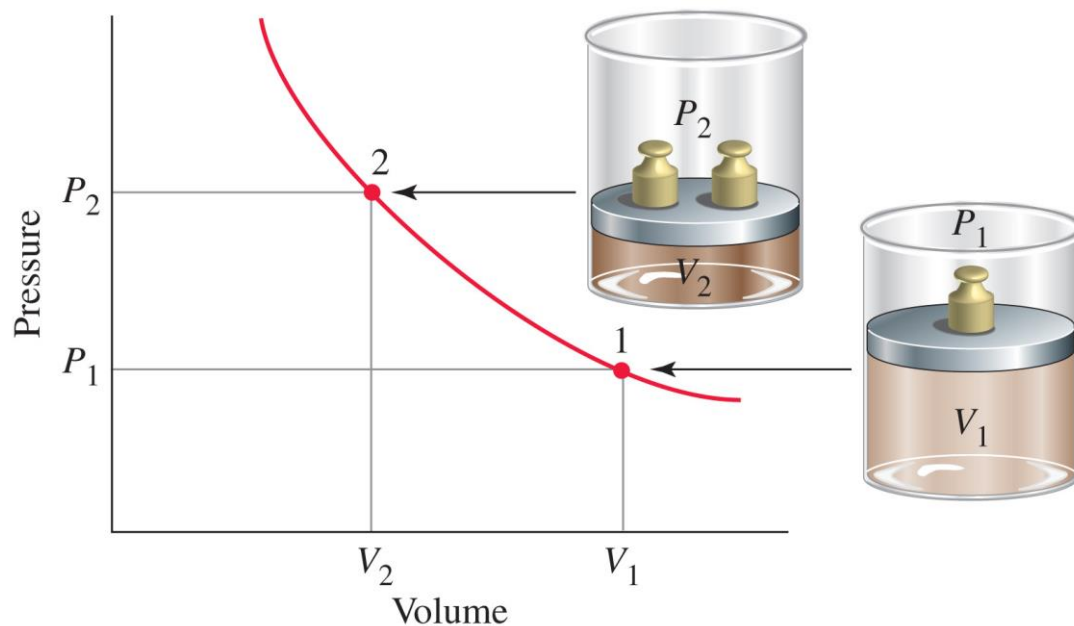
This device is used to measure the difference in pressure between atmospheric pressure and that of a gas in a vessel.

10-3 The Gas Laws

◆ Boyle 1662

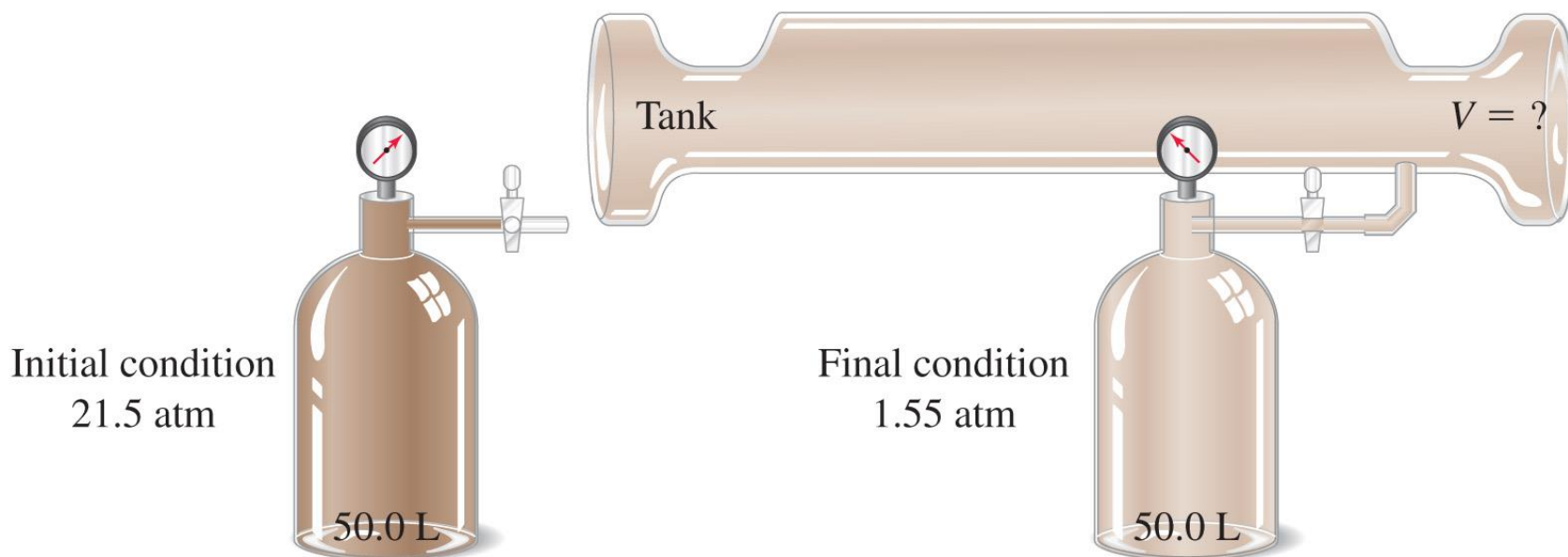
$$P \propto \frac{1}{V}$$

$$PV = \text{constant}$$

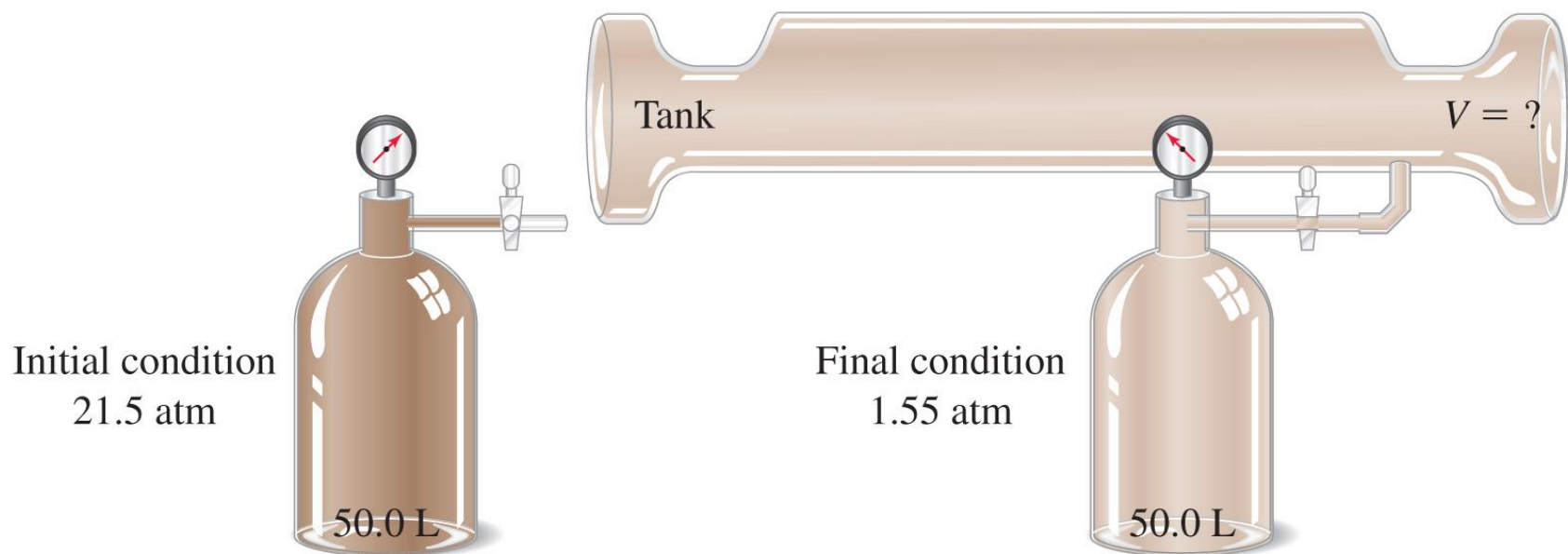


EXAMPLE

Relating Gas Volume and Pressure – Boyle's Law. The volume of a large irregularly shaped, closed tank can be determined. The tank is first evacuated and then connected to a 50.0 L cylinder of compressed nitrogen gas. The gas pressure in the cylinder, originally at 21.5 atm, falls to 1.55 atm after it is connected to the evacuated tank. What is the volume of the tank?



EXAMPLE



$$P_1 V_1 = P_2 V_2 \quad V_2 = \frac{P_1 V_1}{P_2} = 694 \text{ L} \quad V_{\text{tank}} = 644 \text{ L}$$

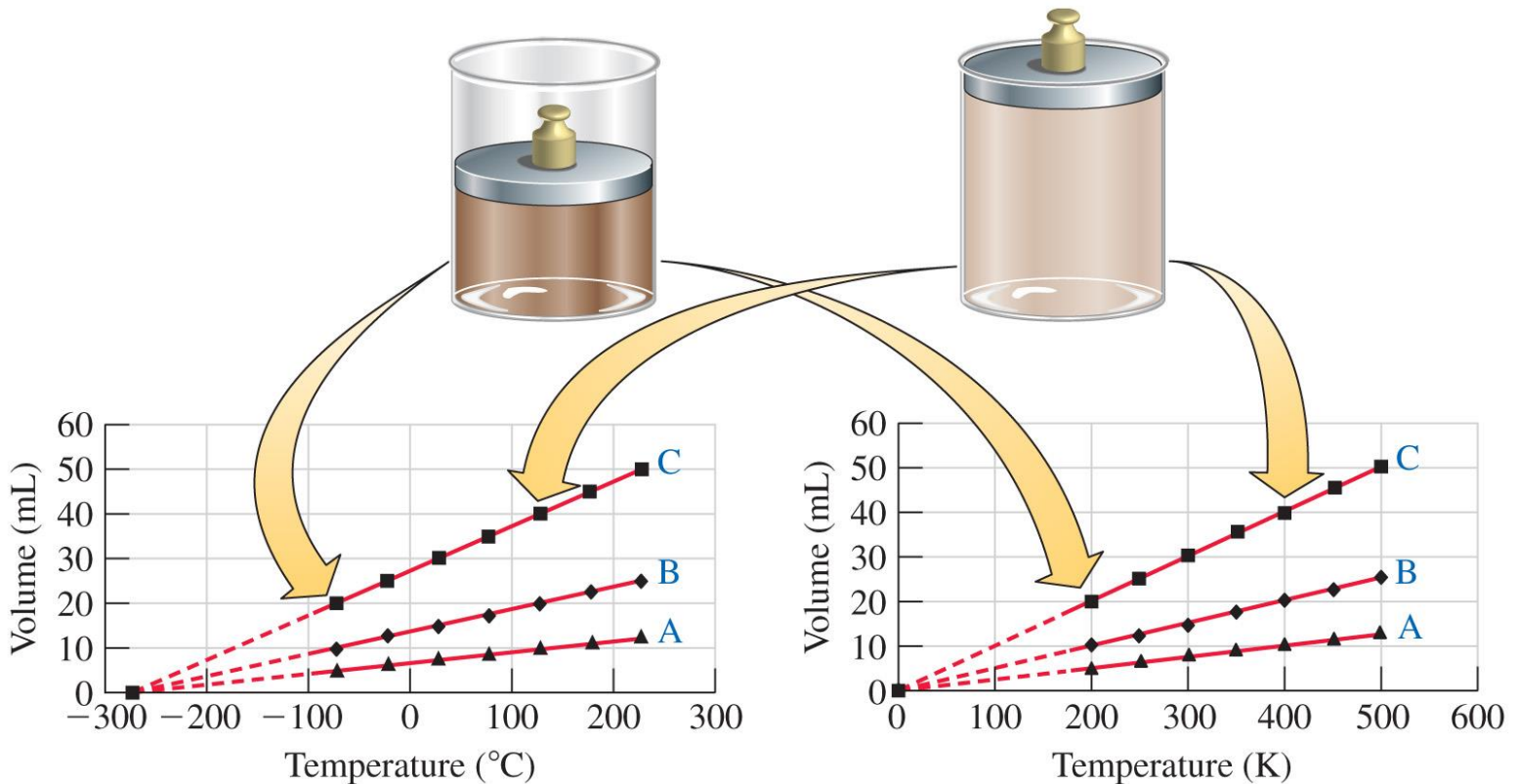
Charles's Law

Charles 1787

$$V \propto T$$

$$V/T = \text{constant}$$

Gay-Lussac 1802



Standard Temperature and Pressure

- ◆ Gas properties depend on conditions.
- ◆ Define standard conditions of temperature and pressure (STP).

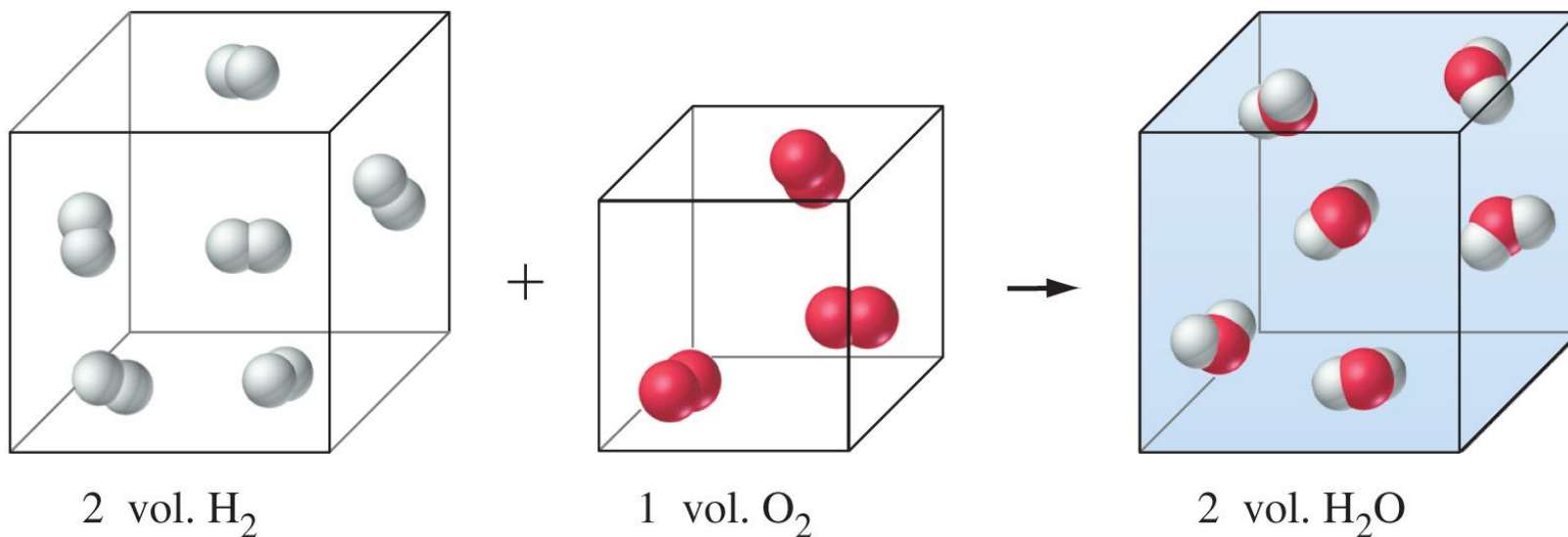
$$P = 1 \text{ atm} = 760 \text{ mm Hg}$$

$$T = 0^{\circ}\text{C} = 273.15 \text{ K}$$

Gay-Lussac's Law

◆ Gay-Lussac 1808

- At a given pressure and temperature, the volumes of gases that react with one another are in the ratios of small whole numbers.



Avogadro's Law

◆ Avogadro's hypothesis

- Equal volumes of gases have equal numbers of molecules *and*
- Gas molecules may break up when they react.

◆ Avogadro's Law

- At an a fixed temperature and pressure:

$$V \propto n \quad \text{or} \quad V/n = \text{constant}$$

Molar volume is the volume of a 1 mol of a gas at STP and equals 22.41 L

10-4 The Ideal Gas Equation

- ◆ Boyle's law $V \propto 1/P$
 - ◆ Charles's law $V \propto T$
 - ◆ Avogadro's law $V \propto n$
- 
- $$V \propto \frac{nT}{P}$$

$$PV = nRT$$

The Gas Constant

$$PV = nRT$$

$$R = \frac{PV}{nT}$$

Units	Numerical Value
L-atm/mol-K	0.08206
J/mol-K*	8.314
cal/mol-K	1.987
m ³ -Pa/mol-K*	8.314
L-torr/mol-K	62.36

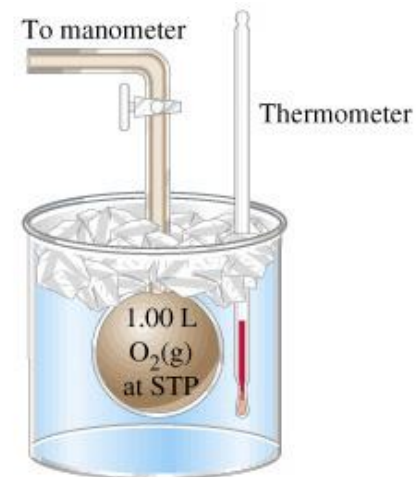
*SI unit

The General Gas Equation

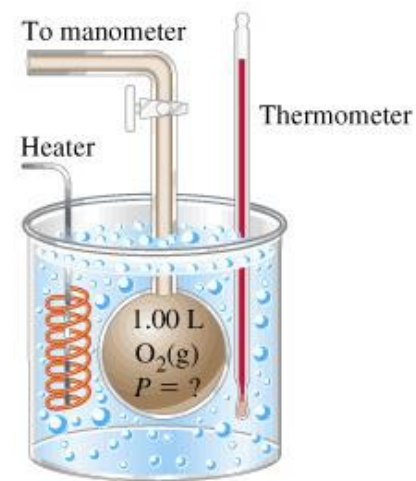
$$R = \frac{P_1 V_1}{n_1 T_1} = \frac{P_2 V_2}{n_2 T_2}$$

If we hold the amount and volume constant:

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

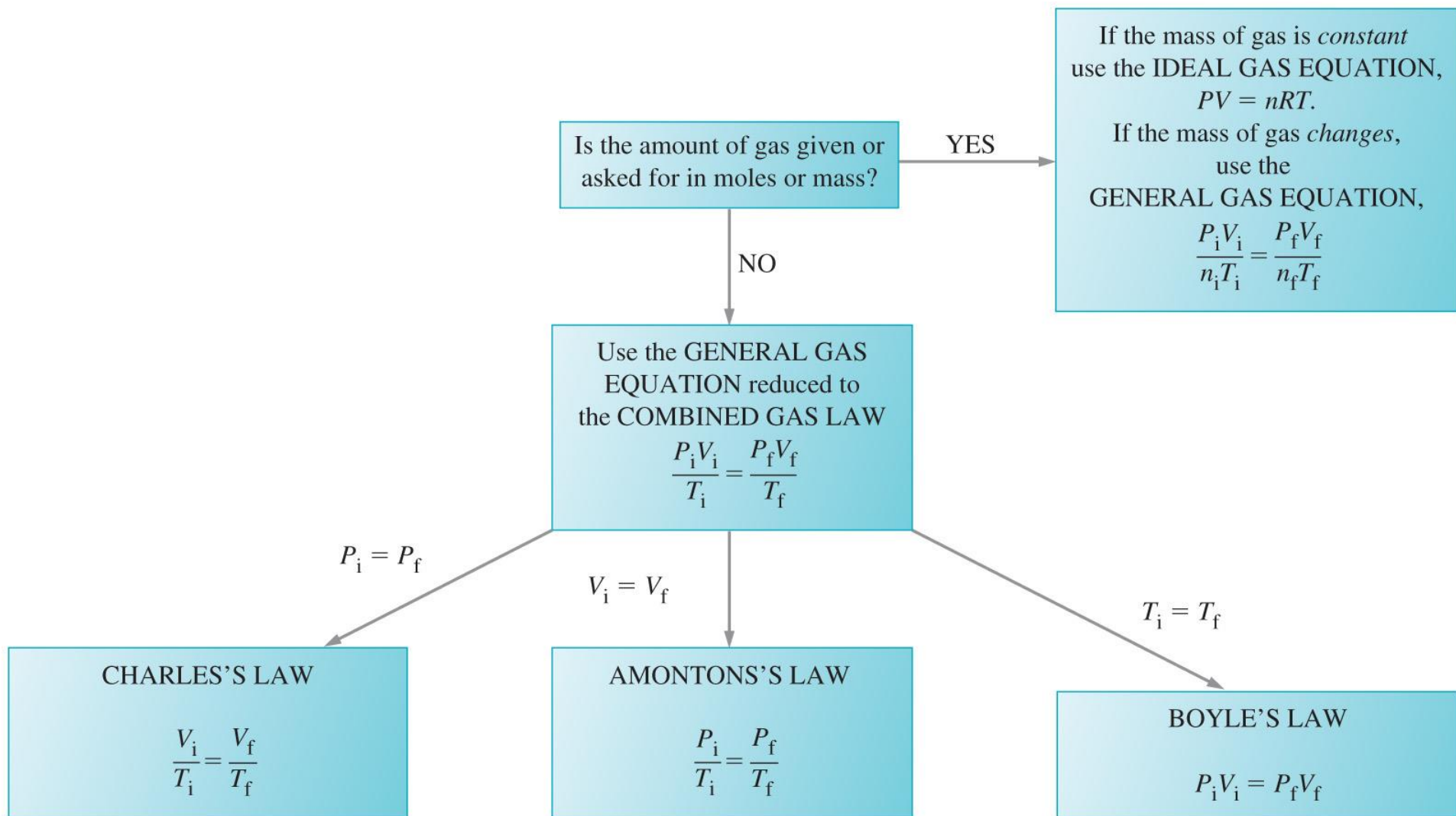


(a) Ice bath



(b) Boiling water

Using the Gas Laws



10-5 Further Applications of the Ideal Gas Equation

Molar Mass Determination

$$PV = nRT \quad \text{and} \quad n = \frac{m}{M}$$

$$PV = \frac{m}{M} RT$$

$$M = \frac{m RT}{PV}$$

EXAMPLE 6-10

Determining a Molar Mass with the Ideal Gas Equation.

Polypropylene is an important commercial chemical. It is used in the synthesis of other organic chemicals and in plastics production. A glass vessel weighs 40.1305 g when clean, dry and evacuated; it weighs 138.2410 when filled with water at 25°C ($\delta_{\text{water}} = 0.9970 \text{ g cm}^{-3}$) and 40.2959 g when filled with propylene gas at 740.3 mm Hg and 24.0°C. What is the molar mass of polypropylene?

Strategy:

Determine V_{flask} . Determine m_{gas} . Use the Gas Equation.

EXAMPLE 6-10

Determine V_{flask} :

$$\begin{aligned} V_{\text{flask}} &= m_{\text{H}_2\text{O}} \div d_{\text{H}_2\text{O}} = (138.2410 \text{ g} - 40.1305 \text{ g}) \div (0.9970 \text{ g cm}^{-3}) \\ &= 98.41 \text{ cm}^3 = 0.09841 \text{ L} \end{aligned}$$

Determine m_{gas} :

$$\begin{aligned} m_{\text{gas}} &= m_{\text{filled}} - m_{\text{empty}} = (40.2959 \text{ g} - 40.1305 \text{ g}) \\ &= 0.1654 \text{ g} \end{aligned}$$

EXAMPLE 5-6

Use the Gas Equation:

$$PV = nRT \quad PV = \frac{m}{M} RT \quad M = \frac{m RT}{PV}$$

$$M = \frac{(0.6145 \text{ g})(0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1})(297.2 \text{ K})}{(0.9741 \text{ atm})(0.09841 \text{ L})}$$

$$M = 42.08 \text{ g/mol}$$

Gas Densities

$$PV = nRT \quad \text{and} \quad d = \frac{m}{V}, \quad n = \frac{m}{M}$$

$$P\textcolor{red}{V} = \frac{\textcolor{red}{m}}{M} RT$$

$$\frac{\textcolor{red}{m}}{\textcolor{red}{V}} = d = \frac{MP}{RT}$$

Gases in Chemical Reactions

- ◆ Stoichiometric factors relate gas quantities to quantities of other reactants or products.
- ◆ Ideal gas equation relates the amount of a gas to volume, temperature and pressure.
- ◆ *Law of combining volumes* can be developed using the gas law.

EXAMPLE 6-12

Using the Ideal gas Equation in Reaction Stoichiometry

Calculations. The decomposition of sodium azide, NaN_3 , at high temperatures produces $\text{N}_2(\text{g})$. Together with the necessary devices to initiate the reaction and trap the sodium metal formed, this reaction is used in **air-bag safety systems**. What volume of $\text{N}_2(\text{g})$, measured at 735 mm Hg and 26°C , is produced when 70.0 g NaN_3 is decomposed?



EXAMPLE 6-12

Determine moles of N_2 :

$$n_{N_2} = 70 \text{ g } \cancel{\text{NaN}_3} \times \frac{1 \text{ mol } \cancel{\text{NaN}_3}}{65.01 \text{ g } \cancel{\text{NaN}_3}} \times \frac{3 \text{ mol } N_2}{2 \text{ mol } \cancel{\text{NaN}_3}} = 1.62 \text{ mol } N_2$$

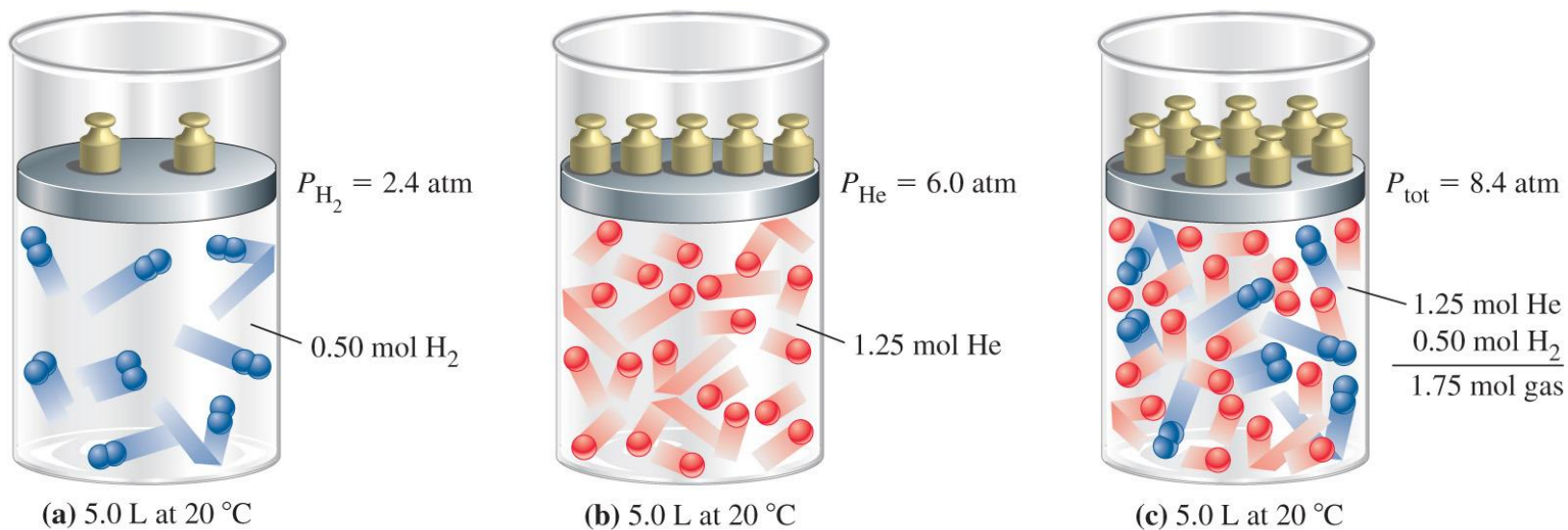
Determine volume of N_2 :

$$V = \frac{nRT}{P} = \frac{(1.62 \text{ mol})(0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1})(299 \text{ K})}{\left\{ (735 \text{ mm Hg}) \times \frac{1.00 \text{ atm}}{760 \text{ mm Hg}} \right\}} = 41.1 \text{ L}$$

10-6 Gas Mixtures and Partial Pressures

- ◆ Gas laws apply to *mixtures* of gases.
- ◆ Simplest approach is to use n_{total} , but....
- ◆ Partial pressure
 - Each component of a gas mixture exerts a pressure that it would exert if it were in the container alone.

Dalton's Law of Partial Pressure



The total pressure of a mixture of gases is the sum of the partial pressures of the components of the mixture.

Partial Pressure

$$P_{\text{tot}} = P_a + P_b + \dots$$

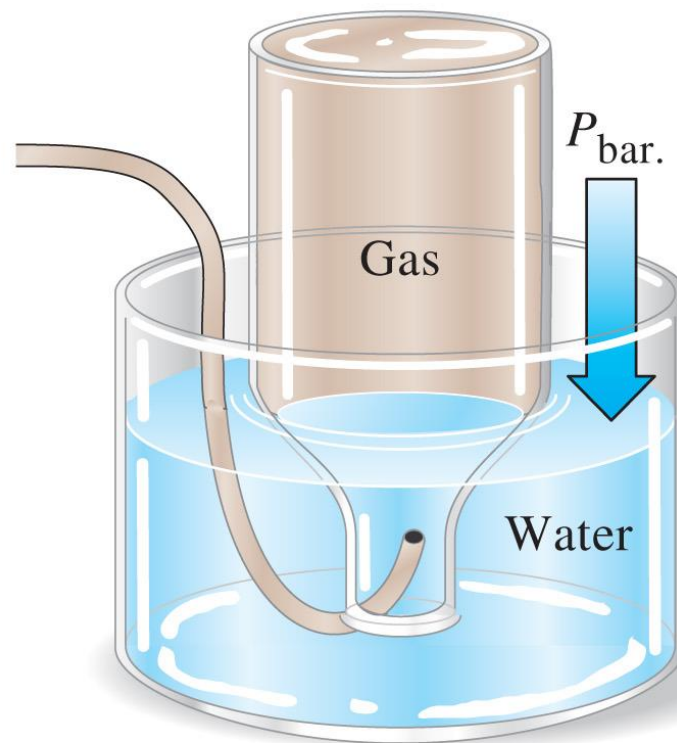
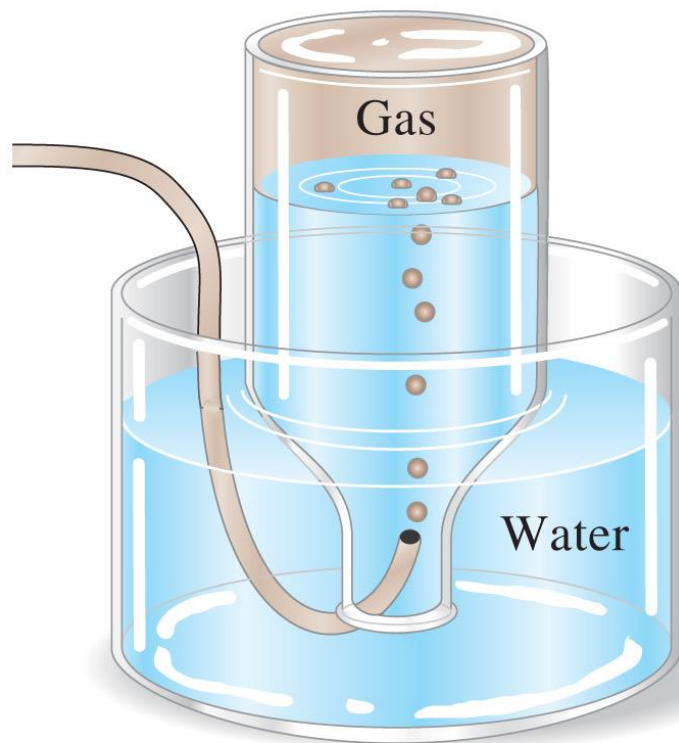
$$V_a = n_a RT / P_{\text{tot}} \quad \text{and} \quad V_{\text{tot}} = V_a + V_b + \dots$$

$$\frac{V_a}{V_{\text{tot}}} = \frac{n_a RT / P_{\text{tot}}}{n_{\text{tot}} RT / P_{\text{tot}}} = \frac{n_a}{n_{\text{tot}}}$$

$$\text{Recall} \quad \frac{n_a}{n_{\text{tot}}} = \chi_a$$

$$\frac{P_a}{P_{\text{tot}}} = \frac{n_a RT / V_{\text{tot}}}{n_{\text{tot}} RT / V_{\text{tot}}} = \frac{n_a}{n_{\text{tot}}}$$

Pneumatic Trough



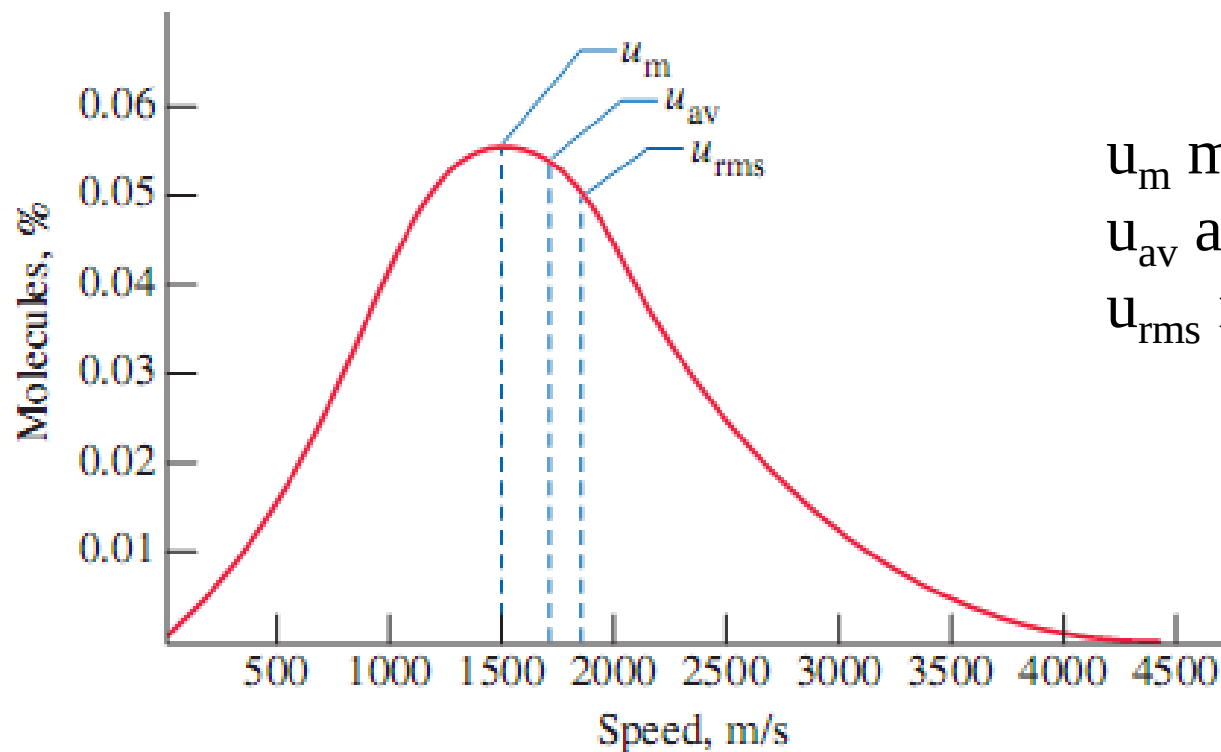
$$P_{\text{tot}} = P_{\text{bar}} = P_{\text{gas}} + P_{\text{H}_2\text{O}}$$

10-7 Kinetic-Molecular Theory of Gases

- ◆ Random motion:
- ◆ Negligible molecular volume
- ◆ Negligible attractive and repulsive forces
- ◆ Constant average kinetic energy
- ◆ Average kinetic energy proportional to temperature



Distributions of Molecular Speed



u_m most probable speed,
 u_{av} average speed
 u_{rms} root mean square speed

$$F(u) = 4\pi \left(\frac{M}{2\pi RT} \right)^{3/2} u^2 e^{-(Mu^2/2RT)}$$

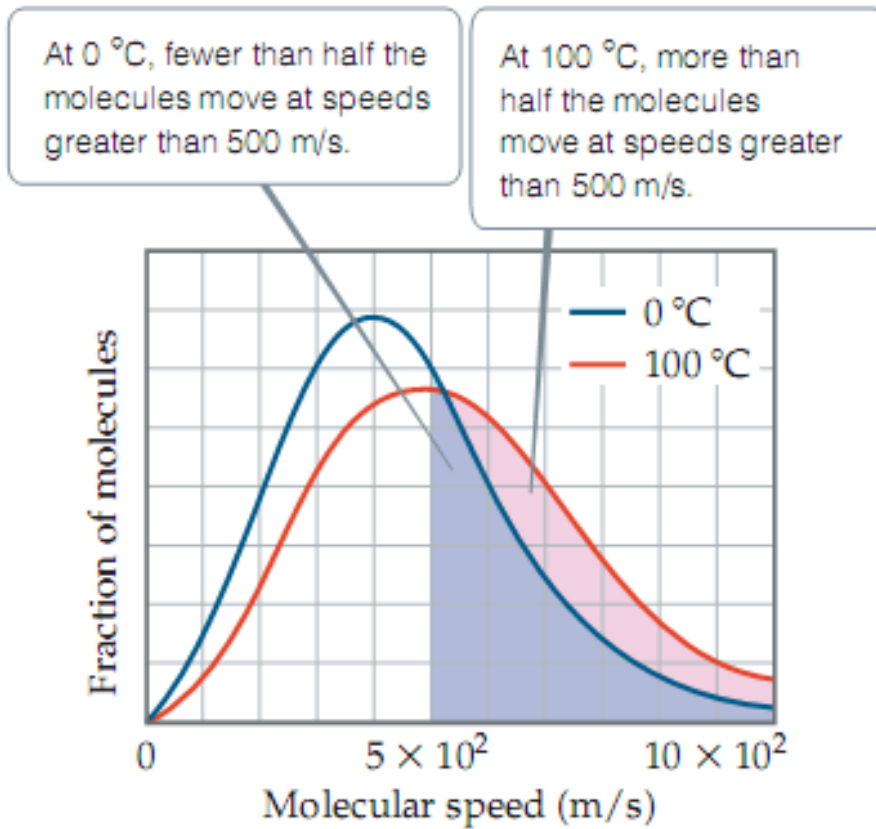
F: fraction of molecules having speed between u and $(u + 1)$ m/s

M: molar mass,

T: absolute temperature

Effects of temperature on molecular speed.

The relative area under the curve for a range of speeds gives the relative fraction of molecules that have those speeds



- ◆ The average kinetic energy of the molecules is proportional to the absolute temperature
- ◆ Energy can be transferred between molecules during collisions, but the average kinetic energy of the molecules does not change with time, as long as the temperature of the gas remains constant.

Distribution of molecular speeds for nitrogen gas

Application of Kinetic-Molecular Theory to the Gas Laws

An increase in volume at constant temperature causes pressure to decrease

A temperature increase at constant volume causes pressure to increase

$P \propto \text{impulse imparted per collision} * \text{collision rate}$

Pressure – Assessing Collision Forces

- ◆ Translational kinetic energy, $e_k = \frac{1}{2} mu^2$
- ◆ Frequency of collisions, $\nu = u \frac{N}{V}$
- ◆ Impulse or momentum transfer, $I = mu$
- ◆ Pressure proportional to impulse times frequency $P \propto \frac{N}{V} mu^2$

$P \propto \text{impulse imparted per collision} * \text{collision rate}$

Pressure

Assume one mole:

$$PV = \frac{1}{3} N_A m \bar{u}^2$$

$PV=RT$ so:

$$3RT = N_A m \bar{u}^2$$

$N_A m = M$:

$$3RT = M \bar{u}^2$$

Rearrange:

$$u_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

Temperature

Modify: $PV = \frac{1}{3} N_A m \bar{u}^2 = \frac{2}{3} N_A \left(\frac{1}{2} m \bar{u}^2 \right)$

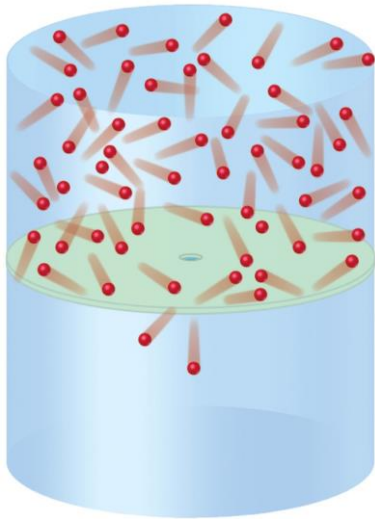
PV=RT so: $RT = \frac{2}{3} N_A \bar{e}_k$

Solve for $\bar{e}_k$: $\bar{e}_k = \frac{3}{2} \frac{R}{N_A} (T)$

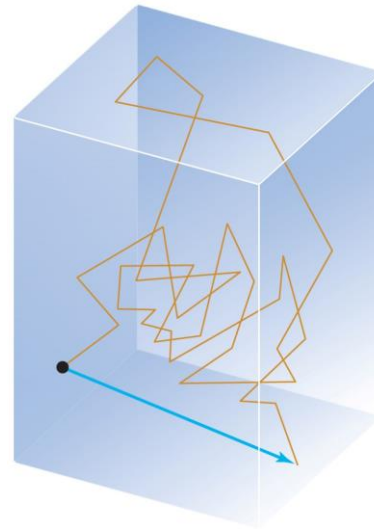
Average kinetic energy is directly proportional to temperature!

10-8 Molecular Effusion and Diffusion

- ◆ Effusion is the escape of gas molecules through a tiny hole into an evacuated space.



- ◆ Diffusion is the spreading of one substance throughout a space or throughout a second substance.



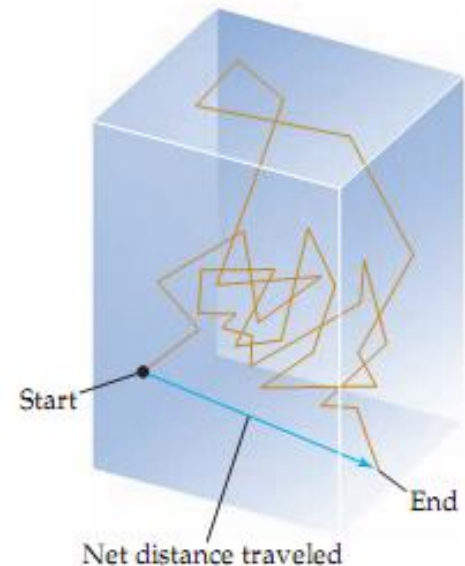
Graham's Law

$$\frac{\text{rate of effusion of A}}{\text{rate of effusion of B}} = \frac{(u_{\text{rms}})_A}{(u_{\text{rms}})_B} = \sqrt{\frac{3RT/M_A}{3RT/M_B}} = \sqrt{\frac{M_B}{M_A}}$$

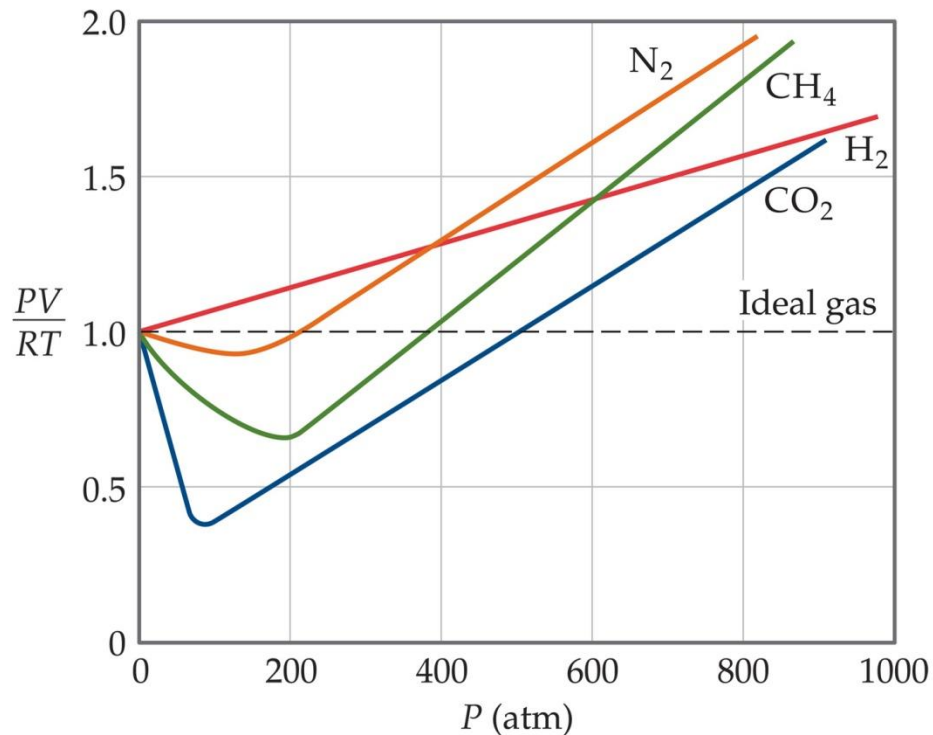
- ◆ Only for gases at **low** pressure (natural escape, not a jet).
- ◆ Tiny orifice (no collisions)
- ◆ Does not apply to diffusion.
- ◆ Ratio used can be:
 - Rate of effusion (as above)
 - Distances traveled by molecules
 - Molecular speeds
 - Amounts of gas effused.
 - Effusion times

Diffusion and Mean Free Path

- ◆ The diffusion of a molecule from one point to another consists of many short, straight-line segments as collisions buffet it around in random directions
- ◆ The average distance traveled by a molecule between collisions, called the molecule's **mean free path**

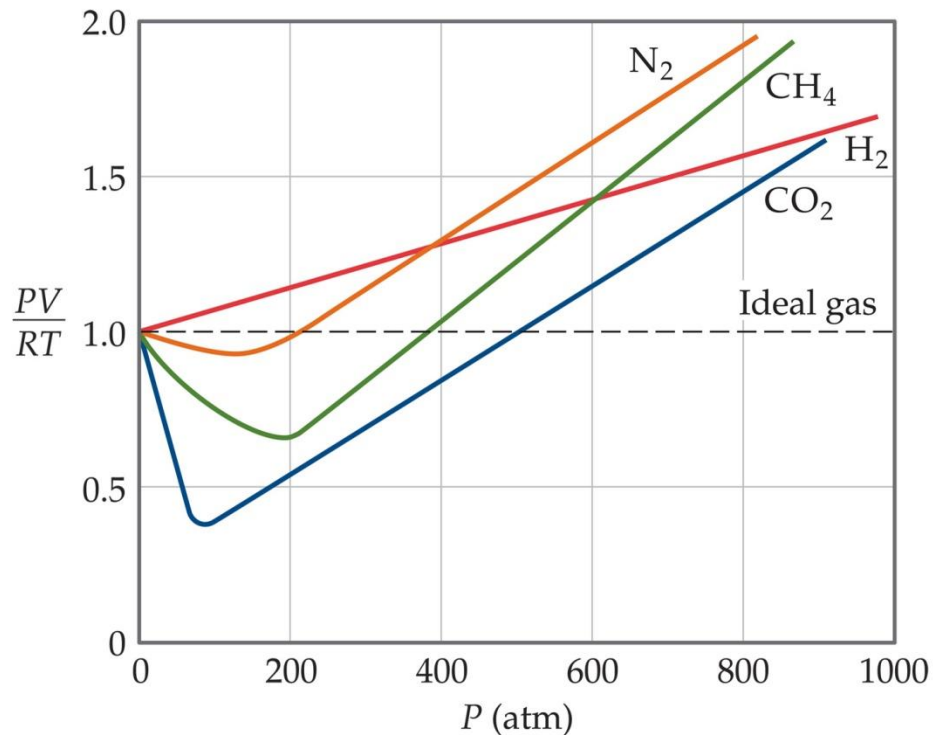


10-9 Real Gases: Deviations from Ideal Behavior



In the real world, the behavior of gases only conforms to the ideal-gas equation at relatively high temperature and low pressure.

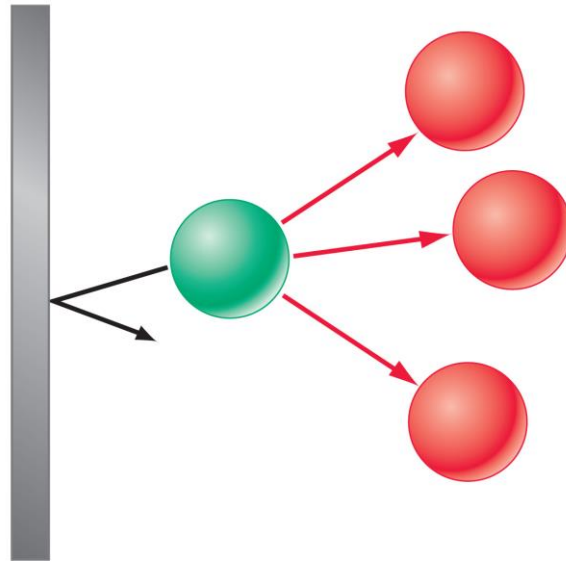
10-9 Real Gases: Deviations from Ideal Behavior



In the real world, the behavior of gases only conforms to the ideal-gas equation at relatively high temperature and low pressure.

10-9 Real Gases: Deviations from Ideal Behavior

- ◆ Compressibility factor $PV/nRT = 1$
- ◆ Deviations occur for real gases.
 - $PV/nRT > 1$ - molecular volume is significant.
 - $PV/nRT < 1$ – intermolecular forces of attraction.



van der Waals Equation

$$\left(P + \frac{n^2 a}{V^2} \right) \left(V - nb \right) = nRT$$

a: a measure of how strongly the gas molecules attract one another,
b: a measure of the finite volume occupied by the molecules

Homeworks

Chemistry *The Central Science*

Fourteenth Edition in SI Units

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All “Practice Exercise 1” of Chapter 10